CUSTOMER SEGMENTATION USING K-MEANS CLUSTERING IN PYTHON

Project Report

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PROBLEM STATEMENT:

To develop an efficient customer segmentation system using K-Means clustering, enabling businesses to understand customer behavior, improve marketing strategies, and boost customer satisfaction.

ABSTRACT:

This project applies unsupervised learning, particularly K-Means clustering, to segment customers based on purchasing behavior. Businesses today collect vast amounts of customer data, but struggle to derive actionable insights. Through clustering, we identify distinct customer groups based on features like spending patterns, age, and income. Visualizations like elbow plots and cluster scatter plots are used to analyze optimal clusters. This project helps businesses design targeted marketing strategies for different customer segments.

OBJECTIVES:

- Load and preprocess customer data.
- Apply K-Means clustering to identify customer segments.
- Determine the optimal number of clusters using the Elbow Method.
- Visualize and interpret customer segments for business insights.

METHODOLOGY:

The customer dataset is imported and cleaned by handling missing values and scaling numerical features. K-Means clustering is applied to the relevant features. The Elbow Method is used to find the optimal number of clusters. Each cluster is analyzed and visualized using scatter plots and pairplots to interpret customer behavior patterns.

SOFTWARE REQUIREMENTS:

- Programming Language: Python 3.x
- Development Environment: Jupyter Notebook / Google Colab / VS Code
- Libraries: pandas, numpy, matplotlib, seaborn, scikit-learn

MODEL IMPLEMENTED:

K-Means Clustering (Unsupervised Machine Learning)

SOURCE CODE (PYTHON):

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler

df = pd.read_csv("Mall_Customers.csv")
X = df.iloc[:, [3, 4]].values

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

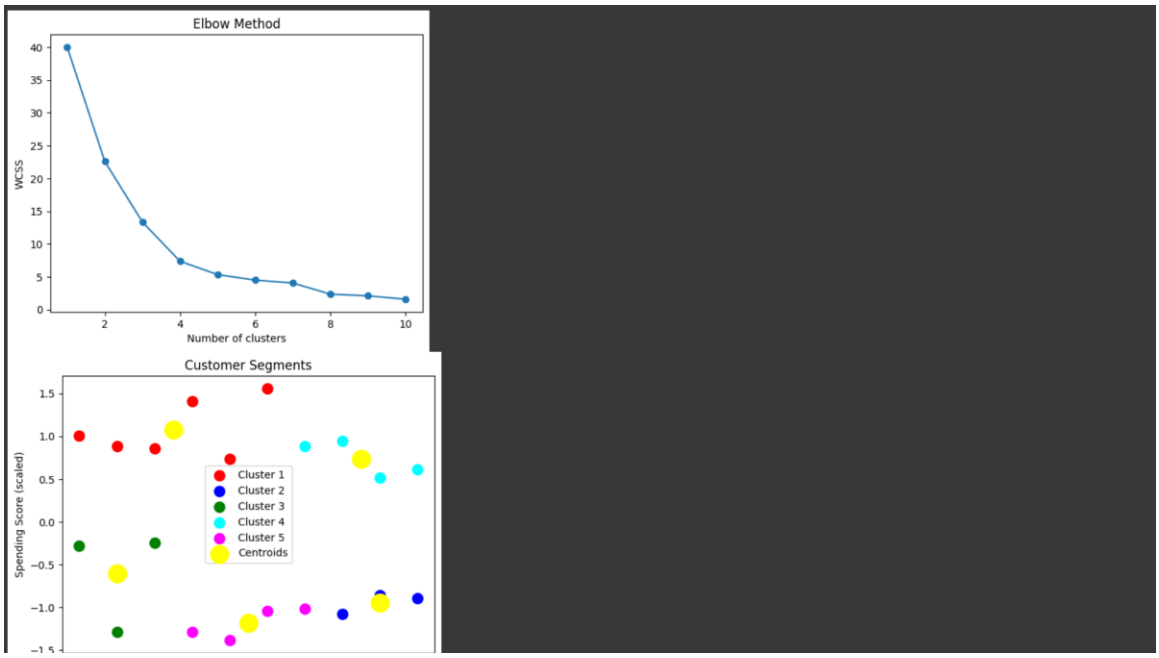
wcss = []
for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

plt.plot(range(1, 11), wcss, marker='o')
plt.title("Elbow Method")
plt.xlabel("Number of clusters")
plt.ylabel("WCSS")
```

```
plt.show()
kmeans = KMeans(n_clusters=5, random_state=42)
y_kmeans = kmeans.fit_predict(X_scaled)
plt.scatter(X_scaled[y_kmeans == 0, 0], X_scaled[y_kmeans ==
0, 1], s=100, c='red', label='Cluster 1')
plt.scatter(X_scaled[y_kmeans == 1, 0], X_scaled[y_kmeans ==
1, 1], s=100, c='blue', label='Cluster 2')
plt.scatter(X_scaled[y_kmeans == 2, 0], X_scaled[y_kmeans ==
2, 1], s=100, c='green', label='Cluster 3')
plt.scatter(X_scaled[y_kmeans == 3, 0], X_scaled[y_kmeans ==
3, 1], s=100, c='cyan', label='Cluster 4')
plt.scatter(X_scaled[y_kmeans == 4, 0], X_scaled[y_kmeans ==
4, 1], s=100, c='magenta', label='Cluster 5')
plt.scatter(kmeans.cluster_centers_[:, 0],
kmeans.cluster_centers_[:, 1], s=300, c='yellow',
label='Centroids')
plt.title("Customer Segments")
plt.xlabel("Annual Income (scaled)")
plt.ylabel("Spending Score (scaled)")
plt.legend()
plt.show()
```

OUTPUT (SAMPLE):

- Elbow Method plot shows minimum WCSS at 5 clusters.
- Scatter plot visualizes 5 distinct customer segments.
- Clusters interpreted as:
  - High income, high spending
  - Low income, low spending
  - Medium income, moderate spending
  - High income, low spending
  - Low income, high spending



RESULTS AND ANALYSIS:

The Elbow Method helped identify 5 as the optimal number of customer segments. Clustering revealed clear patterns of customer behavior that businesses can leverage for targeted marketing. For instance, Cluster 1 represents premium customers, while Cluster 3 represents budget-conscious shoppers.

CONCLUSION:

K-Means clustering provides an effective unsupervised approach for customer segmentation. The model successfully grouped customers into distinct clusters based on purchasing behavior, offering valuable insights for data-driven marketing strategies.

FUTURE ENHANCEMENT:

- Integrate demographic features for richer segmentation.
- Use hierarchical clustering or DBSCAN for comparison.
- Build dashboards for business teams to explore segments interactively.